

Nadia Sultana

nadiasu.08@gmail.com | +8801927085260 | linkedin/sultana-nadia | github/NadiaSu08 | Portfolio

About Me

Prospective PhD student with a research background in Explainable Artificial Intelligence (XAI), Human-Computer Interaction (HCI), and biomedical machine learning. Experienced in designing interpretable deep learning models for healthcare and human-centered systems. Actively seeking PhD opportunities in AI, HCI, and Human-AI Interaction.

Research Interests

Explainable AI (XAI), Human-Computer Interaction (HCI), Human-AI Interaction, Interpretable Machine Learning, Biomedical Signal & Image Analysis, Accessibility-Aware Intelligent Systems, Interactive Data Visualization

Education

M. Sc. in ICT (Ongoing), Institute of Information Technology, Jahangirnagar University Sep 2024 – Present
Current CGPA: 3.71/4.00

B. Sc. in ICT, Institute of Information Technology, Jahangirnagar University Feb 2019 – Sep 2024
CGPA: 3.69/4.00
Honors: Merit-Based Scholarship (for Admission Test, 1st Year, and 3rd Year academic results)

Publications

- S. Akter, M. A. Masud, M. S. I. Promi, N. Sultana et al., “**Dream: A novel explainable neural network for detecting sleep apnea using single-lead ecg signals**,” in *Biomedical Signal Processing and Control (Elsevier, Q1 Journal, 2026)* [Paper]
- M. A. Masud, S. Akter, N. Sultana et al., **MosQNet-SA: Explainable Convolutional-Attention Network for Mosquito Classification with Application as a RESTful API for Dengue and Malaria Risk Mapping**. PLoS ONE [Formally Accepted]
- M. A. Masud, S. Akter, N. Sultana et al., **Multi-Layered Password-Based Steganography: A Novel Approach for Tiered Information Hiding**. In *Proceedings of the 2025 International Conference on Electrical, Computer and Communication Engineering (ECCE), Jan 2025* [Paper]

Manuscripts Under Review

- N. Sultana, S. Afrida, B. Akter et al., **Explainable CNN-GRU Model with Self-Attention for Human Activity Recognition Using Wearable Sensors through Data Augmentation**. IEEE Access (under review)
- M. A. Masud, S. Akter, N. Sultana et al., **Pseudo-RGB data augmentation with deep-learning algorithms for improved Alzheimer’s disease and other disabilities detection**. PLoS ONE (under review)

Research Grants & Fellowships

National Science & Technology (NST) Fellowship | Ministry of Science & Technology, Bangladesh 2025–26
Awarded for M.Sc. research on explainable deep learning for Parkinson’s Disease diagnosis.
Grant Amount: 54,000 BDT

Academic Research Experience

M.Sc. Thesis (Individual Research)

Incep-BiLSTM-Attn: An Explainable Framework for Parkinson’s Disease Diagnosis Using Voice Biomarkers

Developing an explainable deep learning framework using biomedical voice data (IPVS dataset). Designed and implemented a hybrid architecture integrating Inception modules, BiLSTM networks, and attention mechanisms. Applied Explainable AI (XAI) techniques to improve clinical interpretability and transparency.

Supervisor: Dr. Fahima Tabassum, PhD

Professional Experience

Associate UX Engineer, Synesis IT PLC. – Karwan Bazar, Dhaka, Bangladesh Aug 2024 – Present

- Designed data-driven citizen service interfaces for national e-government platforms, improving accessibility and usability.
- Contributed to requirement engineering, user journey mapping, and interface design for systems.
- Developed and customized WordPress-based web platforms, ensuring responsive layouts.
- Created UX flows, prototypes, and interactive interfaces in Figma, aligning design decisions.
- Collaborated with developers to integrate usability research with functional system design.

Technical Skills

- **Programming Languages:** C, C++, Python, Java, C#, MATLAB, SQL
- **Frameworks & Libraries:** TensorFlow, Keras, Scikit-learn, Django, Spring Boot, React, Unity
- **Web & Application Development:** HTML, CSS, JavaScript, Frontend Development, Backend Development, Android Studio
- **Design & Prototyping:** Figma, Adobe Illustrator, WordPress

Training & Certification

Training of Trainers (ToT): B-TopSE Program Agile 2 (Agile Product Development) Dec 2025 – Jan 2026
JICA, BCC, BASIS TCP

- Completed a Training of Trainers program on Agile product development, covering agile methodologies, team facilitation, and agile project management practices.

B-TopSE Program: Software Architecture Course Sep 2025 - Oct 2025
JICA, BCC, BASIS TCP

- Completed intensive training on modern software architecture and system design methodologies, including UML structural modeling and the ICONIX process (Use Case → Domain → Robustness → Design), focusing on scalable system architecture.

Projects

Human Activity Recognition Using ML (UCI-HAR Dataset) github

- Built a system to classify human activities from sensor data using Logistic Regression, SVM, Random Forest, and KNN.
- Performed data preprocessing, feature extraction, and model evaluation.
- Tools: Python, scikit-learn, pandas, NumPy

Predicting Coronary Disease Using Logistic Regression (Framingham Dataset) github

- Analyzed patient data to predict coronary disease risk by performing data preprocessing, feature selection, model training, and evaluation to identify key risk factors using logistic regression.
- Python, scikit-learn, pandas, Seaborn.

 Explore more projects on GitHub

Academic Service & Leadership

Head of Department – Graphic Design, Youth School for Social Entrepreneurs Oct 2021 – Jul 2022

- Led recruitment, supervised daily operations, and managed a team of **40+ members**, contributing to the team's recognition as the **Best Department**.

Vice President (IT), E-Business & Entrepreneurship Club, Jahangirnagar University Aug 2022 – Sep 2024

- Oversaw IT operations, including the club's website and social media, while facilitating networking and improving programs to promote the club's mission.

References

Dr. Mohammad Abu Yousuf

Vice-Chancellor, University of Frontier Technology, Bangladesh

Professor, Institute of Information Technology, Jahangirnagar University, Bangladesh

Email: yousuf@juniv.edu

Mobile: +880 1711-469636